DB Final 2024 - OCR Extracted Text

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fotal Grade: Max. 60 marks = Leconte

No. of pages: 6 pages Final Exam - Model B Time Allowed: 2 Hours

Choose the best suitable answer for each of the following statements:

1- ______is a procedural query language in DBMS, which describes the procedure to obtain the result.

a- SQL b- Relational algebra = c- DDI d- DML

2- The Union operation (U) is

a- commutative b- union compatibility c-a&b d- None of them

3- In Relational Algebra, the Rename operation is denoted by the symbol —

a- 6 b- 1 C- xX d- p

4- _ The operation that is applied to a single relation is called _____ |

a- Unary b- Binary c- Set d- All of them

5- Wecan always combine a cascade of SELECT operations into a single SELECT operation wit __

a- OR b- AND c- NOT d- XOR

6- _____is the inverse operation of the Cartesian product.

a- Join b- Difference c- Division d- Project

7- Normalization of a Database is achieved by following a set of rules called

a- forms b- constraints c- keys d- policies

8- If X is a candidate key of the relation (R), then |

a- R>X b- XR. c- a&b d- None of them

-9- The degree to which the Relation has been normalized refers to ___normal form condition it meets

a- the minimum b- the lowest c- the highest d- All of them

10- A relation with a single-attribute primary key is automatically in at least

a- INF b- 2NF c- 3NF d- 4NF

11- If A->B and B->C are two FDs then A->C is called ___

a- Partial b- Transitive c- Fully d- None

12- If the following functional dependencies in relation R (A, B,C, D)

| AB->C, AC-> D , Then the primary key is

a- A b- B c-A&B d- A&C

13- If the following functional dependencies in relation R (A, B, C, D)

AB ->C, AB ->D, Then the highest normal form of R is

a- INF b- 2NF c- 3NF d- 4NF-

14- If the relation fails to satisfy the SNF condition, then the highest normal form will be _

» ji «a- INF b- 2NF c- 3NF d- 4NF

f every n on-primary-key attribute is fully functionally dependent on the primary key, then the relatior

)6Stéi SON c- 3NF a d-4NF

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—=~

16- DBMS backup the data automatically and it if any data loss happens. |

a- Recover b- copy c- maintain d- retrieve

17- is a group of objects with the same properties, which are identified by the enterprise as having an

independent existence.

a-Entity type b-Attribute type c- Relationship type d-None of them ‘

18- key reference a related table through the primary key of that related table.

a-Primary b-Foreign _c-Candidate d-Super

19- A/An integrity constraint requires the foreign key value to match the primary key value or be ..

null. a eS See 2 eee ee eee ae 2.

a-Primary b-Referential CEnuty eS acd =

20- attributes, are those that consist of a hierarchy of attributes ¥

a-Derived _b-Muti-values c-Composite — d-Simple WC

21- A/An ___ is used to define the overall design of the database. @

a-Database Schema _b-Database Instance c-Code _d-DMC @

22- _____ relationship is one in which a relationship exists between occurrences of the same entity set. @)

a-Unary b-Binary =—_—_—sc-Ternary | d-Bi-directional ()

23. Ina base relation, no attribute of a primary key can be null, refer to integrity. q)

vPrimary __—_b-Referential —c-Entity Special “

24- In EER, the existence of at least one shared subclass leadsto_ i)

aruniom | COON g euctarchy Gc lattice

25- It is a collection of toprams that enable users to create and maintain databases. |

a-DBA _b-ACIS c-DA d-DBMS

26- is the process of minimizing the differences between entities by identifying their

common Characteristics; |

a-Specialization —b- Generalization —_c- Normalization d-Aggregation oa

27- ________are responsible for authorizing access to the database. .

a DBE RK a CDA DML :

28- A virtual relation does not have to be in the database but can be made if a user asks for it at the time of the

TOQUCS Se ee eee may | =

__a-Base relation b-Snapshot. = —c-Query_ d-View ao

29- is the ability of the database to allow multiple users access to the same record without

adversely affecting transaction processing |

a- DB security _b- DB recovery c- DB integrity d-DB Concurrency

30- Mapping the ERD model in Figure 1 produces __relation/s. ,

a-One b-Two c-Three d-Four

31- Mapping the ERD model in Figure 2 produces ___Telation/s. ai

a-One —b-Two c-Three d-Four

32- ER-to-Relational Mapping for Many-to-Many relationship generate

| a-Ignored b- Additional relation — c- additional attribute | d-Merge relations

133-., In ANSI/SPARC model, the schema deals with the user view. a3

a6 Model B

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7 external b-Natural c-Conceptual d-Physical

____ decides how the logical database design is to be physically realized. |

a- DBA b- Data analyst c- DB analyst d- DB designer

35. The number of attributes in a table referred to

a-class b-domain c-cardinality d-degree

36- According to Table 1, the set {A, B} best describes

a-Primary key b-Candidate key c-Super key d-None of them

37- Which SQL clause is used to eliminate duplicate rows from the output? Oe ae

a- ELIMINATE b- UNIQUE c- DELETE d- DISTINCT

38- A database programmer tries to search for employees whose name starts with ‘R’, so he writes the

following query:

SELECT * FROM EMPLOYEE WHERE name = ‘R’; el

Unfortunately, the query is not producing the correct output. What is the correct way to write this query’

a- SELECT * FROM EMPLOYEE HAVING name = *R%";

b- SELECT * FROM EMPLOYEE WHERE name ILIKE ‘R%’;

c- SELECT * FROM EMPLOYEE WHERE name LIKE *%R’;

d- SELECT * FROM EMPLOYEE HAVING name = *%R’;

39- Which SQL query from the following can be used to delete all records of a table without dclcting the

table? ee ee ee oe ke

a- DELETE TABLE table_name; b- DELETE FROM table_name;

c- DROP TABLE table_name; a __ d- DELETE TABLE FROM table_name;

40- Identify the wrong statement about the HAVING clause.

a- HAVING must be used with aGROUP BY clause _

b- Aggregate functions come aftera HAVING clause

c- HAVING clause must be used in pattern matching such as LIKE

d- HAVING clause is used to apply conditions on table rows.

41- Which of the following SQL commands are considered TCL commands?

a- COMMIT and ROLLBACK b- UPDATE and TRUNCATE

__c-SELECTandINSERT ———“s———s—S—S—SC d+ GRANT and REVOKE

42- Which of the following is not a Constraint in SQL? ats

a- PRIMARY KEY b- NOT NULL c-AUTOINCREMENT d- UNION

43- Which operator is used to compare a value to a specified list of values?

a~-ANY b-BETWEEN c-ALL d- IN

44- What operator tests a column for the absence of data? ~

a-NOT b- Exists c- IS NULL d- None

45- If we have not specified ASC or DESC after a SQL ORDER BY clause, the following is used by default.

a- DESC b- ASC c- no default value d-None

46- Which command is used to change the definition of a table in SQL?

a- CREATE b-UPDATE c- ALTER d-SELECT

‘i

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eously-

47 helps multiple users to log into the same database simultan y d- Concurrency

= iSO IT ' bs. J ‘ C- Recovery

a- Security ntegrity

. security —————— =

48- The statements of the security that the system 1s expected to enforce is called 74. = <trictions

a- Rules b- policies c- constraints a

:, reSS ini do. aE

49- refers to the process of determining what a user can GO. “Investigation

~ Authentication b- Protection c- Authorization d- Invesug

50- refers to the accuracy or correctness of the data in database.

i os d- Recovery

a- Integnty b- Security c- Concurrency

Use the attached ‘bank’ DB schema in Figure 3 to solve the following questions:

a- SELECT AVG(amount)FROM loan ORDER BY branch_name,

b- SELECT branch_name, AVG(amount)FROM loan GROUP BY branch_name,

Cc SELECT branch_name, AVG(amount)FROM loan ORDER BY branch_name;,

d- SELECT AVG(amount)FROM loan GROUP BY branch_name,

52- Find all branches with more than one loan. pole _ a

a- SELECT branch_name FROM loan branch_name WHERE COUNT(amount) > |;

b- SELECT branch_name, COUNT(*)FROM loan HAVING COUNT(amount) > |;

c- SELECT branch_name, COUNT(*)FROM loan GROUP BY branch_name WHERE COUNT(amount) > 1;

d- SELECT branch_name, COUNT(*)FROM loan GROUP BY branch_name HAVING COUNT(*) > 1;

53- Find the name of the branch with the smallest number of assets

a- SELECT MIN(assets) FROM branch ;

b- SELECT branch_name FROM branch WHERE assets = (SELECT MIN(assets) FROM branch);

c- SELECT branch_name FROM branch WHERE assets = MIN(assets) :

d- SELECT branch name FROM branch HAVING assets MIN(assets);

54- Findthelargestloanateachbranche

a SELECT branch_name, MAX(amount) FROM loan GROUP BY branch_name;

b- SELECT branch_name FROM loan HAVING MAX(amount) ; __

c- SELECT branch_name FROM loan GROUP BY branch_name WHERE amount =MAX(amount);

d- SELECT branch_name FROM loan WHERE amount = MAX(amount); | |

55- Find customer names who have loans along with branch names from which they borrow.

a- SELECT cust_name, branch_name FROM depositor, account WHERE depositer.acct_id =

account.acct_id; :

b- SELECT cust_name, branch_name FROM depositor OUTER JOIN account ON depositer.acct_id =

account.acct_id; | | a

c- SELECT cust_name, branch_name FROM borrower RIHGT OUTER JOIN loan ON borrower.loan_id

d-SELECT cust name, branch_name FROM borrower JOIN loan ON borrower.loan_id = loan.loan_id —

56- Sl all details of bank accounts in the 'Los Angeles' branch with a balance greater than $250 and less

Rial a6) ECT decid balance FROM account WHERE branch name= Bos Angeles AND (Rae

____ 250 AND < 300); ; s Angeles! ANS ee

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y SELECT acct_id, balance FROM account WHERE branch_name = 'Los Angeles' AND (BALANCE

BETWEEN 250 AND 300);

c- SELECT DISTINCT acct_id, balance FROM account WHERE branch_name = 'Los Angeles' AND

(250 < BALANCE < 300);

AU. d- SELECT * FROM account WHERE branch_name = 'Los Angeles' OR (250 < BALANCE < 300);

te 57- Find all accounts at branches with 'le' somewhere in the name

a- SELECT * FROM account WHERE branch_name LIKE '%le%';

r b- SELECT * FROM account WHERE branch_name = 'le';

F c-SELECT * FROM account WHERE branch_name LIKE '%le_';

d- SELECT * FROM account WHERE branch_name LIKE ' le%';

58- To use acct_id as a foreign key in the depositor table, you should add which statement of the following to

depositor schema?

a- REFERENCE account(acct_id);

b- FOREIGN KEY REFERENCES account(acct_id);

c- FOREIGN KEY REFERENCES account(acct_id);

d- FOREIGN KEY (account_id) CHAR(10) REFERENCES account(acct_id);

59- Retrieve a list of all bank branch details from "branch" table, ordered by branch city, with cach citys

branches listed in reverse order of holdings.

a- SELECT * FROM branch ORDER BY branch_city, assets ASC;

b- SELECT * FROM branch ORDER BY branch_city DESC, assets ASC;

c- SELECT * FROM branch ORDER BY branch_city ASC, assets DESC;

d- SELECT * FROM branch ORDER BY assets DESC;

60- Find the number of branches that currently have loans.

a- SELECT loan_idFROMloam;

b- SELECT COUNT(DISTINCT branch_name) FROM loan;

| ¢- SELECT COUNT(branch_name) FROM loan;

d- SELECT COUNT(branch_name) FROM loan HAVING amount;

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—— | Employee < Has , Computer'

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| Student WwW

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Model B

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A301 New York 360 Joann = AT

A-307 | Seattle 716 Srrvth AR

A318 | LosAngeles | 860 Reryralds | Mone

A319 | New York 80 Lows =| ASOT

A322 | LosAngeles | 275 Reynolds | AW

account depositor

L421 San Francisco | 7500 Anderson =§- | 497

L445 | Los Angeles —_| 2000 Jackson = | L-419 |

L437 Las Vegas 4300 Lewis [471 |

L419 | Seattle 2900 Smith L445 |

loan borrower |

Brighton Brooktyn 700000 |

Downtown Brooklyn 9000000 |

Redwood Pato Alto 21000000